



Term Evaluation (Odd) Semester Examination September 2025

Roll no. 2294038

Name of the Course: B.Tech

Semester: VII

Name of the Paper: Advanced Computer Architecture

Paper Code: TCS 704

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1.

(10 Marks)

a.

CO3

A CPU runs at 2 GHz with a CPI of 2 and executes a program with 10^9 instructions.

- I. Calculate execution time.
- II. If the clock speed is doubled to 4 GHz but CPI remains the same, calculate the new execution time.
- III. Now assume 30% of the instructions suffer a memory stall of 50 ns (unchanged with faster clock). Recalculate the execution time and compare.
- IV. Why is the expected performance gain not achieved? Relate this to the Myopic View of Computer Architecture.

OR

b.

CO3

A CPU has a two-level cache hierarchy:

- L1 cache hit time = 1 ns, hit rate = 90%
- L2 cache hit time = 10 ns, hit rate = 95% (for remaining 10%)

Main memory access time = 100 ns

(i) Calculate the average memory access time (AMAT). (ii) If L1 hit rate improves to 95%, recalculate AMAT. Comment on the performance improvement.

Q2.

(10 Marks) CO1

- a. Differentiate between temporal locality and spatial locality. Illustrate each with an example. For a loop that repeatedly accesses A[5], explain how temporal locality helps.

OR

b.

CO1

What is Amdahl's Law? A program spends 40% of its time in a parallelizable section. Using Amdahl's Law:

- I. Calculate speedup for 4 processors.
- II. Calculate speedup for 8 processors.



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III. Comment on diminishing returns.

Q3.

(10 Marks)

a.

CO2

Classify parallel computers. For each type, give a numerical example:

- If a task of size 1000 operations is executed, estimate execution time assuming each architecture processes operations at different parallelism levels (you may assume SISD = serial, SIMD = 4 ops/cycle, MISD = redundant execution, MIMD = 4 processors).

OR

b.

CO2, CO3

A dual-core system shares a cache. One core writes a value $X = 5$ into cache, while the other core reads X from memory as 3.

- Explain the cache coherence problem.
- Illustrate with an example how the MESI protocol (or any one protocol) solves this issue.

Q4.

(10 Marks)

a.

CO1

Moore's Law states transistor count doubles approximately every 18–24 months.

- If a processor had 1 million transistors in 1980, estimate transistor count in 2000.
- Discuss why physical scaling has slowed in modern era and its implications.

OR

b.

CO2

Compare direct-mapped, fully associative, and set-associative cache organizations.

If you have 16 blocks of main memory and 4 cache lines, show how block 6 would be placed in each organization.

Q5.

(10 Marks)

a.

CO1



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Define dependability in computer architecture.

- I. If a system has $MTBF = 200$ hours and $MTTR = 2$ hours, calculate availability.

Discuss how energy, power, and cost constraints affect system dependability.

OR

b.

CO3

Explain the principle of Virtual Memory.

- I. A system has 32-bit virtual addresses and 4 KB page size. Calculate:
a. Number of pages in virtual address space.
b. Number of offset bits.
- II. Explain how paging helps efficient memory utilization.